Chapter 8. Baire Spaces and Dimension Theory

The Baire category theorem, nowhere-differentiable functions, covering dimension and embeddings

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8.1 Baire spaces

Motivation. A complete metric space cannot be a countable union of nowhere-dense sets. This simple statement, the Baire category theorem, is a powerful existence tool: to show that some object exists, show that the set of "bad" objects is small in the sense of category.

$A \subset X$ has **empty interior** if it contains no nonempty open set. X is a **Baire space** if every countable union of closed sets with empty interior has empty interior.

Example 8.1

$\mathbb{Q}$ is not a Baire space: it is the countable union of its one-point sets, which are closed with empty interior. $\mathbb{Z}_+$ is a Baire space: every nonempty subset is open, so the only set with empty interior is $\emptyset$. The irrationals form a Baire space, since they admit a complete metric (they are homeomorphic to $\mathbb{Z}_+^\omega$); we do not prove this here.

Lemma 8.1 (48.1)

X is a Baire space iff for every countable collection $\{U_n\}$ of dense open sets, $\bigcap U_n$ is dense.

Proof. U is open and dense iff $X - U$ is closed with empty interior, and $\bigcap U_n$ is dense iff $X - \bigcap U_n = \bigcup (X - U_n)$ has empty interior. ■

Theorem 8.2 (48.2, Baire category theorem)

If X is a compact Hausdorff space or a complete metric space, then X is a Baire space.

Proof. Let A_n be closed with empty interior and U_0 nonempty open; we find a point of U_0 outside every A_n. Since $U_0 \not\subset A_1$, pick $y \in U_0 - A_1$. X is regular (Theorem 4.8, or metric), so there is a neighbourhood U_1 of y with $\overline{U_1} \cap A_1 = \emptyset$ and $\overline{U_1} \subset U_0$; in the metric case choose also $\operatorname{diam} U_1 < 1$. Inductively, given nonempty open U_{n-1}, choose nonempty open U_n with $\overline{U_n} \subset U_{n-1}$, $\overline{U_n} \cap A_n = \emptyset$, and (metric case) $\operatorname{diam} U_n < 1/n$. The nested nonempty closed sets $\overline{U_n}$ have nonempty intersection: by the finite intersection property in the compact case, and by Lemma 8.3 in the complete case. A point of the intersection lies in U_0 and in no A_n. ■

Lemma 8.3 (48.3)

If $C_1 \supset C_2 \supset \cdots$ are nonempty closed sets in a complete metric space with $\operatorname{diam} C_n \to 0$, then $\bigcap C_n \neq \emptyset$.

Proof. Pick $x_n \in C_n$. For $n, m \geq N$, both lie in C_N, so $d(x_n, x_m) \leq \operatorname{diam} C_N$; the sequence is Cauchy and converges to some x. For each k, $x_n \in C_k$ for $n \geq k$, so $x \in C_k$. ■

Lemma 8.4 (48.4)

An open subspace Y of a Baire space X is a Baire space.

Proof. Let A_n be closed in Y with empty interior in Y, and $\overline{A_n}$ its closure in X, so $\overline{A_n} \cap Y = A_n$. If U is a nonempty open subset of X contained in $\overline{A_n}$, then U meets $A_n \subset Y$, and $U \cap Y$ is a nonempty open subset of Y contained in A_n, impossible. So $\overline{A_n}$ has empty interior in X. If U is open in Y (hence in X) and $U \subset \bigcup A_n \subset \bigcup \overline{A_n}$, then $U = \emptyset$ because X is Baire. ■

Theorem 8.5 (48.5)

Let X be a Baire space, (Y,d) metric, and $f_n : X \to Y$ continuous with $f_n(x) \to f(x)$ for every x. Then the set of points where f is continuous is dense in X.

Proof. For $N \in \mathbb{Z}_+$ and $\epsilon > 0$ let $A_N(\epsilon) = \{x : d(f_n(x), f_m(x)) \le \epsilon \ \forall n, m \ge N\}$, closed as an intersection of closed sets.

Step 1. every nonempty open U contains a nonempty open V such that, for some N, $d(f_n(x), f(x)) \le \epsilon$ for all $x \in V$, $n \ge N$. Indeed, since $(f_n(x))$ is Cauchy for each x, $X = \bigcup_N A_N(\epsilon)$; U is Baire (Lemma 8.4), so some $A_N(\epsilon) \cap U$ has nonempty interior V in U, and V is open in X. Letting $m \to \infty$ in the defining inequality gives the claim.

Step 2. let G_M be the union of all open V as in Step 1 for $\epsilon = 1/M$. It is open and, by Step 1, dense. So $C = \bigcap_M G_M$ is dense (Lemma 8.1). Let $x_0 \in C$ and $\epsilon > 0$; choose M with $1/M < \epsilon/3$, an open $V \ni x_0$ and N with $d(f_n(x), f(x)) \le 1/M$ on V for $n \ge N$, and a neighbourhood $W \subset V$ of x_0 with $d(f_N(x), f_N(x_0)) < \epsilon/3$ on W. Then for $x \in W$,

$$d(f(x), f(x_0)) \le d(f(x), f_N(x)) + d(f_N(x), f_N(x_0)) + d(f_N(x_0), f(x_0)) < \epsilon.$$

So f is continuous at each point of the dense set C. ∎

8.2 A nowhere-differentiable function

Motivation. As an application of Baire's theorem, most continuous functions (in the sense of category) are nowhere differentiable, even though no explicit example is needed in the proof.

Theorem 8.6 (49.1)

Let $h : [0,1] \to \mathbb{R}$ be continuous and $\epsilon > 0$. There is a continuous, nowhere differentiable $g : [0,1] \to \mathbb{R}$ with $|h(x) - g(x)| < \epsilon$ for all x.

Proof. Let $I = [0,1]$ and $\mathcal{C} = \mathcal{C}(I, \mathbb{R})$ with the sup metric ρ; it is complete (Theorem 7.6), hence Baire. For $f \in \mathcal{C}$, $x \in I$ and $0 < k \le \tfrac{1}{2}$ let

$$\Delta f(x,k) = \max\Big\{\Big|\frac{f(x+k) - f(x)}{k}\Big|, \Big|\frac{f(x-k) - f(x)}{k}\Big|\Big\},$$

the maximum taken over those of $x \pm k$ lying in I (at least one does), and $\|\Delta_k f\| = \inf_x \Delta f(x,k)$. Let U_n be the set of f for which $\|\Delta_k f\| > n$ for some $k \le 1/n$.

(a) $\bigcap U_n$ consists of nowhere differentiable functions. If $f \in \bigcap U_n$, there are $k_n \le 1/n$ with $\Delta f(x, k_n) > n$ for all x. If f were differentiable at x (one-sidedly at an endpoint), the difference quotients would tend to $f'(x)$, so $\Delta f(x,k_n) \to |f'(x)|$, contradicting $\Delta f(x,k_n) > n$.

(b) U_n is open. Let $f \in U_n$ with $M = \|\Delta_k f\| > n$, and $\delta = (M-n)k/4$. If $\rho(f,g) < \delta$, then $|g(x \pm k) - g(x)| \ge |f(x \pm k) - f(x)| - 2\delta$, so $\Delta g(x,k) \ge \Delta f(x,k) - (M-n)/2$ and $\|\Delta_k g\| \ge (M+n)/2 > n$.

(c) U_n is dense. Let $f \in \mathcal{C}$ and $\epsilon' > 0$. By uniform continuity choose a partition of I on whose subintervals f varies by less than $\epsilon'/2$; the piecewise linear p interpolating f at the partition points satisfies $\rho(f,p) < \epsilon'/2$ ($p(x)$ is a convex combination of two values each within $\epsilon'/2$ of $f(x)$). Let L bound the slopes of p, so $|p(x) - p(y)| \le L|x-y|$. Choose m with $r = m\epsilon'/4 \ge L + n + 1$, and let ϕ be the sawtooth that is linear on each $[j/m, (j+1)/m]$, alternately rising from 0 to $\epsilon'/4$ and falling back, so $0 \le \phi \le \epsilon'/4$ and ϕ has slope $\pm r$. Put $g = p + \phi$, so $\rho(f,g) < \epsilon'$, and $k = \min(1/n, 1/2m)$. Each x lies in some $[j/m, (j+1)/m]$ of length $1/m \ge 2k$, so $x+k$ or $x-k$ lies in the same subinterval,

where $|\phi(x \pm k) - \phi(x)| = rk$. Then $|g(x \pm k) - g(x)| \geq rk - Lk$, so $\Delta g(x,k) \geq r - L \geq n+1$ for all x, and $g \in U_n$.

By Lemma 8.1, $\bigcap U_n$ is dense, so it contains some g with $\rho(h,g) < \epsilon$. ■

8.3 Introduction to dimension theory

Motivation. What does it mean for a space to have dimension m? The covering definition says: every open cover can be refined so that no point lies in more than $m+1$ sets. With it we show that every compact m-manifold embeds in $\mathbb{R}^{2m+1}$.

A collection $\mathcal{A}$ has **order** $m+1$ if some point lies in $m+1$ elements and no point lies in more than $m+1$. X is **finite dimensional** if for some m every open covering has an open refinement covering X of order at most $m+1$; the least such m is the **topological dimension** $\dim X$. It is a topological invariant. Every compact subspace of $\mathbb{R}$ has dimension at most 1; an arc has dimension 1.

Theorem 8.7 (50.1)

If X is finite dimensional and $Y \subset X$ is closed, then Y is finite dimensional and $\dim Y \leq \dim X$.

Proof. Let $m = \dim X$ and $\mathcal{A}$ a covering of Y by sets open in Y; choose A' open in X with $A' \cap Y = A$. Cover X by the A' and $X - Y$, take an open refinement $\mathcal{B}$ of order at most $m+1$ covering X, and intersect its elements with Y: this refines $\mathcal{A}$, covers Y, and has order at most $m+1$. ■

Theorem 8.8 (50.2)

If $X = Y \cup Z$ with Y, Z closed and finite dimensional, then $\dim X = \max\{\dim Y, \dim Z\}$.

Proof. Let $m = \max\{\dim Y, \dim Z\}$; $\dim X \geq m$ by Theorem 8.7.

Step 1. every open covering $\mathcal{A}$ of X has an open refinement covering X in which each point of Y lies in at most $m+1$ elements. Take a refinement $\mathcal{C}$ of $\{A \cap Y\}$ by sets open in Y, covering Y, of order at most $m+1$. For $C \in \mathcal{C}$ choose U_C open in X with $U_C \cap Y = C$ and $A_C \in \mathcal{A}$ with $C \subset A_C$. The collection $\{U_C \cap A_C\} \cup \{A - Y : A \in \mathcal{A}\}$ works: it covers X, refines $\mathcal{A}$, is open since Y is closed, and a point $y \in Y$ in $U_C \cap A_C$ lies in C.

Step 2. given $\mathcal{A}$, apply Step 1 to get $\mathcal{B}$ (good at points of Y), then Step 1 for Z to $\mathcal{B}$ to get $\mathcal{D}$ (good at points of Z). Choose $f : \mathcal{D} \to \mathcal{B}$ with $D \subset f(D)$ and let $E(B) = \bigcup\{D : f(D) = B\}$. The sets $E(B)$ are open, cover X and refine $\mathcal{A}$. If x lies in distinct $E(B_1), \ldots, E(B_k)$, then x lies in distinct $B_1, \ldots, B_k$ and in distinct $D_1, \ldots, D_k$ with $f(D_i) = B_i$; so $k \leq m+1$ whether $x \in Y$ or $x \in Z$. ■

Corollary 8.9 (50.3)

If $X = Y_1 \cup \cdots \cup Y_k$ with each Y_i closed and finite dimensional, then $\dim X = \max\{\dim Y_1, \ldots, \dim Y_k\}$.

Proof. Induction on k using Theorem 8.8. ■

Points $x_0, \ldots, x_k$ of $\mathbb{R}^N$ are **geometrically independent** if $\sum a_i x_i = 0$ and $\sum a_i = 0$ imply all $a_i = 0$; equivalently, no x_j lies in the plane P spanned by the others, $P = \{\sum t_i x_i : \sum t_i = 1\}$. A set

of points of $\mathbb{R}^N$ is in **general position** if each subset of at most $N+1$ of its points is geometrically independent.

Lemma 8.10 (50.4)

Given $x_1, \dots, x_n \in \mathbb{R}^N$ and $\delta > 0$, there are $y_1, \dots, y_n$ in general position with $|x_i - y_i| < \delta$.

Proof. Induction: let $y_1 = x_1$, and suppose $y_1, \dots, y_p$ are in general position. Each subset T of at most N of them spans a plane $P(T)$ of dimension at most $N-1$, a closed set with empty interior. A finite union of closed sets with empty interior has empty interior (if $U \subset A \cup B$ is open and nonempty, then $U - A$ is a nonempty open subset of B). So choose y_{p+1} within δ of x_{p+1} outside every $P(T)$. A subset S of at most $N+1$ of $y_1, \dots, y_{p+1}$ is independent: either $S \subset \{y_1, \dots, y_p\}$, or $S = T \cup \{y_{p+1}\}$; then if $\sum_T a_i y_i + a y_{p+1} = 0$ with $\sum a_i + a = 0$ and $a \neq 0$, $y_{p+1} = \sum(-a_i/a) y_i \in P(T)$, a contradiction; so $a = 0$ and all $a_i = 0$. ∎

Theorem 8.11 (50.5, imbedding theorem)

Every compact metrizable space X of topological dimension m imbeds in $\mathbb{R}^{2m+1}$.

Proof. Let $N = 2m+1$, $|\cdot|$ the square metric on $\mathbb{R}^N$, and $\mathcal{C}(X, \mathbb{R}^N)$ with the sup metric ρ, complete and hence Baire. For continuous f let $\Delta(f) = \sup\{\operatorname{diam} f^{-1}(z) : z \in f(X)\}$ and $U_\epsilon = \{f : \Delta(f) < \epsilon\}$.

U_ϵ is open: let $\Delta(f) < b < \epsilon$. The set $A = \{(x, y) : d(x, y) \geq b\}$ is compact and $|f(x) - f(y)| > 0$ on it, with positive minimum δ (if $A \neq \emptyset$). If $\rho(f, g) < \delta/2$, then $g(x) \neq g(y)$ on A, so $\Delta(g) \leq b < \epsilon$.

U_ϵ is dense: let f and $\delta > 0$ be given. Cover X by open sets of diameter less than $\epsilon/2$ whose images under f have diameter less than $\delta/2$; by compactness and $\dim X = m$ there is a finite open refinement $\{U_1, \dots, U_n\}$ covering X of order at most $m+1$, and a partition of unity $\{\phi_i\}$ dominated by it (Theorem 4.16). Choose $x_i \in U_i$ and, by Lemma 8.10, points z_i in general position with $|z_i - f(x_i)| < \delta/2$. Let $g = \sum \phi_i z_i$. If $\phi_i(x) > 0$, then $x \in U_i$ and $|f(x) - z_i| < \delta$; since $f(x) - g(x) = \sum \phi_i(x)(f(x) - z_i)$, $|f(x) - g(x)| < \delta$, and by compactness $\rho(f, g) < \delta$. If $g(x) = g(y)$, then $\sum(\phi_i(x) - \phi_i(y)) z_i = 0$ with coefficients summing to 0, and only indices with $x \in U_i$ or $y \in U_i$ appear, at most $2(m+1) = N+1$ of them. By general position all coefficients vanish. Choosing i with $\phi_i(x) > 0$ gives $\phi_i(y) > 0$, so $x, y \in U_i$ and $d(x, y) < \epsilon/2$. Hence $\Delta(g) \leq \epsilon/2$ and $g \in U_\epsilon$.

By Lemma 8.1, $\bigcap_n U_{1/n}$ is dense, hence nonempty; a map g in it is injective, hence an imbedding (Theorem 3.20). ∎

Theorem 8.12 (50.6)

Every compact subspace of $\mathbb{R}^N$ has topological dimension at most N.

Proof.

Step 1. an open covering of $\mathbb{R}^N$ by sets of square-metric diameter at most 1, of order at most $N+1$. The cubical cells of $\mathbb{R}^N$ are the products $\sigma = \prod I_i$ with each I_i either $\{k_i\}$ ($k_i \in \mathbb{Z}$) or $(k_i, k_i + 1)$; let $P(\sigma)$ be the set of indices of the first kind. Every point lies in exactly one cell. Write $t_i(y)$ for the distance from y_i to $\mathbb{Z}$, and let

$$V(\sigma) = \Big\{y : \ |y_i - k_i| < \tfrac{1}{2} \ (i \in P(\sigma)), \quad y_i \in (k_i, k_i + 1) \ (i \notin P(\sigma)), \quad \max_{i \in P(\sigma)} t_i(y) < \min_{i \notin P(\sigma)} t_i(y)\Big\},$$

an open set of diameter at most 1 containing σ. If $\sigma \neq \tau$ and $|P(\sigma)| = |P(\tau)|$, then $V(\sigma) \cap V(\tau) = \emptyset$: if $P(\sigma) = P(\tau)$ some k_i differs, which is impossible on a point coordinate (nearest integer) and on an interval coordinate (disjoint intervals); otherwise there are $i \in P(\sigma) - P(\tau)$ and $j \in P(\tau) - P(\sigma)$,

and a common point would satisfy $t_i < t_j$ and $t_j < t_i$. So each point lies in at most one $V(\sigma)$ for each value $|P(\sigma)| \in \{0, \dots, N\}$, hence in at most $N+1$ of them.

Step 2. let $X \subset \mathbb{R}^N$ be compact, and let $\mathcal{A}$ be an open covering of X with Lebesgue number δ (Lemma 3.28). Choose $0 < s < \delta$; the sets $sV(\sigma) \cap X$ have diameter less than δ, so each lies in an element of $\mathcal{A}$; they form an open refinement covering X of order at most $N+1$. ■

Corollary 8.13 (50.7)

Every compact m-manifold has topological dimension at most m.

Proof. Each x has an open U with a homeomorphism $h : U \to W \subset \mathbb{R}^m$ open. Choose a closed ball $\overline{B} \subset W$ about $h(x)$; then $C_x = h^{-1}(\overline{B})$ is compact (so closed in X), contains the neighbourhood $h^{-1}(B)$ of x, and has dimension at most m by Theorem 8.12. Finitely many C_x cover X; apply Corollary 8.9. ■

Corollary 8.14 (50.8)

Every compact m-manifold imbeds in $\mathbb{R}^{2m+1}$.

Proof. It is compact Hausdorff, hence regular, with a countable basis, so metrizable (Theorem 4.12); its dimension k is at most m, and it imbeds in $\mathbb{R}^{2k+1} \subset \mathbb{R}^{2m+1}$ by Theorem 8.11. ■

Corollary 8.15 (50.9)

A compact metrizable space imbeds in some $\mathbb{R}^N$ iff it has finite topological dimension.

Proof. ($\Rightarrow$) Theorem 8.12.

($\Leftarrow$) Theorem 8.11. ■

Without compactness: every m-manifold, and more generally every metrizable space of dimension m with a countable basis, imbeds in $\mathbb{R}^{2m+1}$ (Menger–Nöbeling).

8.4 Selected exercises

Motivation. These exercises from Munkres widen the Baire category theorem to locally compact spaces, use it to show that no function on $\mathbb{R}$ is continuous exactly at the rationals, and check that the covering dimension sees connectedness.

Exercise 8.1 (§48, Ex. 3; locally compact spaces are Baire)

Show that every locally compact Hausdorff space X is a Baire space.

Solution. Let $A_1, A_2, \dots$ be closed with empty interior and U_0 nonempty open. We show $U_0 \not\subset \bigcup_n A_n$.

Step 1. Since A_1 has empty interior, $U_0 - A_1$ is a nonempty open set. Pick a point in it; by Theorem 3.35 it has a neighbourhood U_1 with $\overline{U_1}$ compact and $\overline{U_1} \subset U_0 - A_1$. Inductively, given a nonempty open U_{n-1}, choose a nonempty open U_n with $\overline{U_n}$ compact and $\overline{U_n} \subset U_{n-1} - A_n$.

Step 2. The sets $\overline{U_n}$ are nonempty, closed in the compact space $\overline{U_1}$, and nested, so they have the finite intersection property and $\bigcap_n \overline{U_n} \neq \emptyset$ (Theorem 3.23).

Step 3. A point x of the intersection lies in $\overline{U_1} \subset U_0$, and for each n, $x \in \overline{U_n} \subset U_{n-1} - A_n$, so $x \notin A_n$. Hence $U_0 \not\subset \bigcup A_n$: the union has empty interior. ■

Exercise 8.2 (§48, Ex. 7; continuity exactly on a countable dense set)

Let D be a countable dense subset of $\mathbb{R}$. Show that there is no $f : \mathbb{R} \to \mathbb{R}$ that is continuous precisely at the points of D.
(a) Show that the set C of points at which a function $f : \mathbb{R} \to \mathbb{R}$ is continuous is a G_δ set.
(b) Show that D is not a G_δ set.

Solution.

(a) For $U \subset \mathbb{R}$ let $\operatorname{diam} f(U) = \sup\{|f(s) - f(t)| : s, t \in U\}$, and let U_n be the union of all open U with $\operatorname{diam} f(U) < 1/n$; U_n is open.

Step 1. $C \subset \bigcap U_n$: if f is continuous at x, choose $\delta > 0$ with $|f(y) - f(x)| < 1/(3n)$ for $|y - x| < \delta$. Then $U = (x - \delta, x + \delta)$ has $\operatorname{diam} f(U) \le 2/(3n) < 1/n$, so $x \in U_n$.

Step 2. $\bigcap U_n \subset C$: let $x \in \bigcap U_n$ and $\epsilon > 0$; choose n with $1/n < \epsilon$. Then x lies in an open U with $\operatorname{diam} f(U) < 1/n$, so $|f(y) - f(x)| < \epsilon$ for all $y \in U$. Thus f is continuous at x, and $C = \bigcap_n U_n$.

(b) Suppose $D = \bigcap_n W_n$ with W_n open. Each $W_n \supset D$ is dense, and for $d \in D$ the open set $V_d = \mathbb{R} - \{d\}$ is dense. The family $\{W_n\} \cup \{V_d : d \in D\}$ is countable, and

$$\bigcap_n W_n \cap \bigcap_{d \in D} V_d = D - D = \emptyset.$$

But $\mathbb{R}$ is a complete metric space, hence a Baire space (Theorem 8.2), so this countable intersection of dense open sets is dense by Lemma 8.1, a contradiction.

By (a) and (b), the set of continuity points of f is never D. ■

Exercise 8.3 (§50, Ex. 2; connected spaces have dimension at least 1)

Show that a connected T_1 space X with more than one point has dimension at least 1.

Solution. If X is not finite dimensional there is nothing to prove, so suppose $\dim X = 0$ and derive a contradiction.

Step 1. Choose $x \neq y$. As X is T_1, $\mathcal{A} = \{X - \{x\}, X - \{y\}\}$ is an open covering. Since $\dim X = 0$, it has an open refinement $\mathcal{B}$ covering X of order at most 1, i.e. the elements of $\mathcal{B}$ are pairwise disjoint.

Step 2. Let $V \in \mathcal{B}$ contain x. V lies in an element of $\mathcal{A}$, which must be $X - \{y\}$ since $x \in V$; so $y \notin V$. Let W be the union of the other elements of $\mathcal{B}$.

Step 3. V and W are open, disjoint, cover X, and are nonempty ($x \in V$, $y \in W$). This is a separation of the connected space X, a contradiction. ■